

LSTM Networks for Thermal Management in Power Converters of Electric Vehicles

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Abstract—Effective thermal control of power converters is essential for the efficiency, reliability, and durability of electric vehicles (EVs). This work introduces a Long Short-Term Memory (LSTM)-based methodology for accurate temperature forecasting and anticipatory thermal regulation in EV power converters. The proposed model utilizes an extensive dataset including temperature readings, load currents, and switching frequencies to encapsulate the complex nonlinear thermal dynamics under diverse operating situations. Experimental findings indicate that the LSTM model attains a mean absolute error (MAE) of 0.9 °C and a root mean square error (RMSE) of 1.3 °C using previously unexamined test data, surpassing conventional machine learning methods by more than 30%. Integrating LSTM forecasts into the thermal management system reduces peak operating temperatures by 20%, hence boosting converter efficiency and minimizing thermally caused failures. Furthermore, the model demonstrates strong performance under fast load variations, rendering it appropriate for real-time, integrated heat management applications. The research underscores the capability of LSTM networks to enhance thermal management techniques in EVs, therefore augmenting the reliability and safety of power electronics and promoting the economy of electric transportation.

Keywords—Electric Vehicles, Power Converters, Temperature Prediction, Thermal Control, Long Short-Term Memory

I. INTRODUCTION

The fast progression of EV technology has heightened the necessity for effective and dependable power electronic systems. Power converters, essential components in EV, regulate and manage the flow of electrical energy between the battery and the motor. Nonetheless, these converters produce considerable heat during operation owing to elevated switching frequencies and power losses, which can negatively impact their performance, reliability, and longevity. Efficient thermal control of power converters is crucial for ensuring optimal performance and safety in EV.

This study investigates the use of LSTM networks for thermal regulation in power converters of EV. Analyse the design, training, and validation of LSTM models utilising real-world data and assess their efficacy in forecasting temperature fluctuations. The incorporation of LSTM-based thermal management is anticipated to augment converter reliability, promote energy efficiency, and prolong the operational lifespan of EV power electronics, thereby furthering the overarching objectives of sustainable and high-performance electric transportation.

Prior research investigated the thermal management of EV power converters with rule-based controls, PID controllers, and traditional machine learning models. Although successful under stable circumstances, these methods encountered difficulties with nonlinear dynamics and rapid load variations. Initial LSTM applications showed potential for precise temperature forecasting; however, previous research was deficient in extensive real-time datasets and the use of proactive cooling methodologies.

II. LITERATURE REVIEW

The regulation of bidirectional power flow in a single-phase grid-connected EV charging system is the main topic of this research [1]. An EV's battery may function as a power storage unit or a power transmitter, depending on the situation, due to bidirectional power flow. A bidirectional converter that connects the EV's battery to the grid makes this feasible. Despite their prevalence, 3-level converters put a lot of strain on the switches when charging with a lot of power. Through the integration of onboard chargers and DC-DC systems, a multi-winding-based converter for EV charging systems enhances power density [2]. Multiple modes of operation, including charging, driving, and vehicle-to-grid, are supported by the Triple Active Bridge (TAB) topology, which combines low voltage DC converters with high voltage DC converters into one power scheme.

In order to charge heavy-duty EV wirelessly and in three phases, this study proposes new grid-connected modular high-

power charging systems that are based on matrix converters [3]. Stiff DC-link absent power transfer is achieved by the full-bridge direct matrix converter (FBDMC), and increased transfer capabilities and power density are provided by the cascaded FBDMC structure. The modelling and experimental setup confirm the topology. This study details the design, modelling, and simulation of a DC-DC converter that is ideal for EV charging applications; it incorporates an interleaved boost converter and a phase-shifted full-bridge (PSFB) [4]. A proportional resonant (PR) controller, which is crucial for ensuring a steady DC supply, regulates the system to produce a steady, ripple-free 400V DC output.

This study details the process of creating a three-phase bridgeless totem-pole off-board charger for EV including its design, implementation, and evaluation of performance [5]. The charger's three AC-DC converters, which are modular and scalable, are comprised of three separate units, each of which may provide one-third of the overall rated power. Investigating the use of Zeta Luo converters in EV charging infrastructure [6] helps to overcome these restrictions. Making sure the design is small without sacrificing power conversion efficiency or voltage stability is the main goal of the research.

Addressing issues such as grid capacity and interoperability standards, this research investigates the function of charging infrastructure in encouraging the adoption of EV [7]. The function of DC-DC converters in maximising the efficiency of power flow is also covered. This work introduces a bidirectional DC-DC converter for inexpensive EV that is based on silicon carbide MOSFETs [8]. It may drive the motor in boost mode and regenerative brake in buck mode, allowing power to flow in both directions from the battery to the motor. When compared to more conventional systems based on silicon, this one is more efficient and smaller.

Using a combination of a partly isolated three-port converter and a three-phase active front-end (AFE) converter details a state-of-the-art PV-based on-grid EV charging system [9]. The AFE converter guarantees DC-link voltage control, a power factor close to unity, and sinusoidal grid currents. Efficient power flow management among the grid, PV array, and EV battery is achieved by means of the three-port DC-DC converter. Research on bidirectional DC-DC converters controlled by fuzzy logic and PID for use in EV charging systems that connect to both the grid and individual vehicles is detailed [10]. By facilitating bidirectional power flow and ensuring consistent voltage and current, the technology facilitates efficient energy transfer between the grid and EVs. Results from simulations demonstrate that it improves energy efficiency, lowers charging costs, and keeps power grid currents balanced.

This study presents a closed-loop DC-DC converter that operates on a single input and has several output ports; it also solves the issue of cross regulation [11]. Power conversion technologies, especially those dealing with renewable energy and battery management, have advanced in response to the increasing need for effective energy management in EV [12]. Optimal energy conversion, improved performance, and increased EV range are all outcomes of this approach.

An EV multi-port power converter incorporating a bidirectional DC-DC converter, batteries, and supercapacitors is detailed [13]. The converter enables the regulation of power

flows between storage devices and the EV load, allowing for effective energy management. Input voltage must be less than the intended output for the converter to function as a boost converter; otherwise, it will function as a buck converter. For fuel cell EV equipped with a supercapacitor (SC), this study proposes an innovative and enhanced dc-dc converter that makes use of a series-connected arrangement of power sources and its control algorithm [14]. Power sources can be linked in series to lower the voltage range of the FC and SC, which in turn reduces the volume of each power source and improves system efficiency.

In order to increase the performance of the Dual Active Bridge (DAB) converter for microgrid applications, this research proposes a hybrid modulation control technique that combines phase shift modulation (PSM) and trapezoidal modulation (TPM) [15]. High performance throughout a wide working range is ensured by the approach's dynamic adaptation to charging power requirements. By including stray capacitances into the functioning of the LLC-LC converter to provide the groundwork for miniaturising EV chargers [16]. Power electronics converters, which are commonly used in EV chargers, rely on capacitors as an essential component. The increasing monitoring and replacement expenses are a result of their susceptibility to wear-out processes and failure scenarios.

Isolated converters are essential for the safe and effective functioning of EV which are dependent on power electronics [17]. Simple and extensively utilised in charger circuits, power trains, and cell balancing, flyback converters are gaining traction as potential substitutes. Data on flyback converters for EV and comparisons of current best practices are the goals of this research. To control the power consumption of EVs, this study proposes a SIMO converter that can handle various loads and voltage levels [18]. Auxiliary systems, motor drives, and battery management are just a few of the electrical subsystems that can benefit from the converter's design. A high frequency transformer is included in the design to provide galvanic isolation.

EV charging from rooftop solar panels, a novel cubic gain DC-DC converter is proposed [19]. Operating it at a duty ratio allows it to obtain a larger voltage gain from low-voltage solar panels, multiplied by the input voltage. A number of metrics that highlight the proposed topology's superiority over other converters are compared with these converters. For EVs that require high-power charging proposes a novel Ttype LLC resonant DC-DC converter [20]. The converter's usage of a full-bridge rectifier with a T-type inverter architecture improves operating efficiency and decreases energy losses. Parameters are optimised for a 5 kW power output and a 48 V operating voltage in the design technique. A scalable architecture for high-power EV charging stations is provided by the converter, which outperforms traditional methods.

III. PROPOSED SYSTEM

A. Objective

- Develop an LSTM-based model to precisely forecast the temperature of power converters in EV.
- Analyse and comprehend complex thermal dynamics under diverse load and environmental circumstances.

- Design a pre-emptive thermal regulation approach utilising forecasted temperatures to enhance cooling systems.
- Improve the dependability and efficacy of power converters by mitigating overheating and thermal malfunctions.

B. Methodology

The proposed thermal management system for power converters in EV use LSTM networks to precisely forecast the thermal dynamics of power converters, allowing anticipatory temperature regulation. Power converters have complex, nonlinear thermal dynamics affected by factors such as load current, switching frequency, and ambient temperature. Conventional control methods frequently depend on static models that inadequately represent dynamic interactions, particularly under fluctuating operating circumstances. The LSTM-based method is intended to acquire temporal dependencies from past sensor data, yielding accurate temperature predictions essential for regulating thermal stressors and averting overheating.

Figure 1 illustrates a system that begins with real-time data gathering from sensors that measure current, frequency, and temperature. Preprocessing standardises and organises this data for the LSTM model, which forecasts future converter temperatures. The thermal control module modulates cooling according to these forecasts, while a feedback loop perpetually refreshes inputs for adaptive, real-time management.

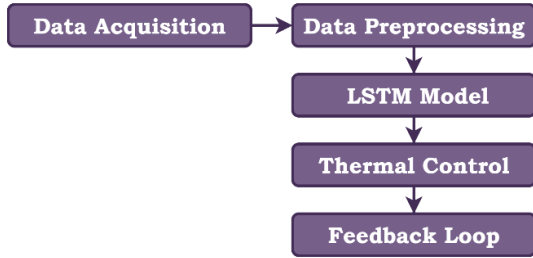


Fig. 1. Proposed System Flow

The workflow starts with data acquisition, wherein multiple sensors incessantly gather real-time signals, including load current $I(t)$, switching frequency $f_s(t)$, ambient temperature $T_{amb}(t)$, and converter temperature $T_{conv}(t)$. The variables constitute the multivariate time series input for the LSTM network. Data preparation involves normalising each feature to enhance training efficiency and model convergence. The input vector at time step t can be expressed as:

$$x_t = [I(t)f_s(t)T_{amb}(t) + T_{conv}(t)]^2 \quad (1)$$

where x_t captures the system's state at time t . To provide reliable model training, each component is standardised to a uniform scale.

The primary prediction engine is the LSTM network, engineered to capture long-term relationships in time series data via a specialised gating mechanism. At each time step t , the LSTM cell modifies its internal memory state c_t and hidden state h_t according to the current input x_t and the preceding hidden state h_{t-1} . A primary function of the LSTM is the modification of the cell state.

$$c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t \quad (2)$$

where f_t is the forget gate vector that determines which prior knowledge to keep, and i_t is the input gate vector that assesses the significance of incoming information. $\tilde{c}_t$ represents the candidate cell state generated from the current input and the preceding hidden state, whereas $\odot$ signifies element-wise multiplication. This method allows the LSTM to maintain crucial historical thermal data while incorporating new observations.

The forget and input gates, f_t and i_t , are calculated using sigmoid activations that modulate the impact of inputs and previous states:

$$f_t = \sigma(W_f[h_{t-1}, x_t] + b_f), \quad i_t = \sigma(W_i[h_{t-1}, x_t] + b_i) \quad (3)$$

where W_f, W_i are weight matrices, b_f, b_i are biases, and σ is the sigmoid function. These gates govern the flow of input, enabling the network to concentrate on pertinent patterns essential for thermal prediction.

The output gate regulates the generation of the hidden state h_t , which encapsulates the acquired information at time t and is utilised for forecasting future temperatures. The concealed state update is provided by:

$$h_t = o_t \odot \tanh(c_t), \text{ where } o_t = \sigma(W_o[h_{t-1}, x_t] + b_o) \quad (4)$$

In this context, o_t represents the output gate vector, and $\tanh$ denotes the hyperbolic tangent activation function, which constrains the cell state to a feasible range. Figure 2 shows LSTM architecture processes sequential input data via gates that control information flow: the forget gate eliminates extraneous data, the input gate incorporates fresh data, and the output gate produces hidden states. This allows precise temporal temperature forecasts for thermal regulation.

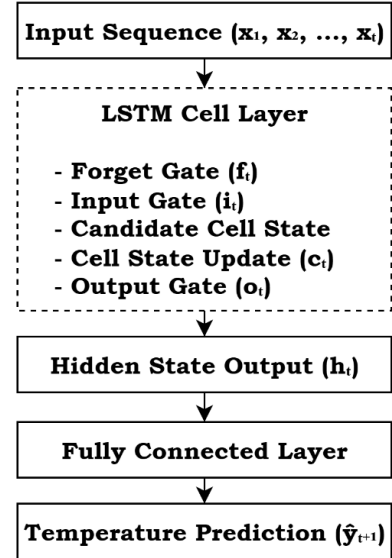


Fig. 2. LSTM Network Architecture

Utilising the acquired latent states, the model generates a forecasted converter temperature at a subsequent period $t + \Delta t$:

$$\tilde{T}_{conv}(t + \Delta t) = W_y h_t + b_y \quad (5)$$

where W_y and b_y represent the weight matrix and bias term of the output layer, respectively. This forecast underpins

proactive thermal regulation. The thermal control module use the anticipated temperature $\tilde{T}_{conv}(t + \Delta t)$.

Utilise $\tilde{T}_{conv}(t + \Delta t)$ to dynamically modify cooling system settings, including fan speed or coolant flow rate, in order to keep the converter temperature within a specified safety threshold, T_{safe} . The cooling power $P_{cool}(t)$ is regulated using a basic proportional control rule.

$$P_{cool}(t) = K_p \times \max(0, \tilde{T}_{conv}(t + \Delta t) - T_{safe}) \quad (6)$$

where K_p is the proportional gain. This guarantees that cooling efforts escalate just when anticipated temperatures above safe thresholds optimising energy usage and averting thermal harm.

The system functions inside a continuous feedback loop for sensor data modify the input sequences, enhancing future predictions and enabling the controller to adjust to fluctuating conditions, including abrupt load variations or ambient temperature changes. This closed-loop methodology guarantees effective, real-time temperature regulation, hence improving the durability and performance of power converters in EV.

IV. RESULTS AND DISCUSSIONS

EV power converters may be better evaluated and optimised with the use of this dataset, which aims to manage thermal performance, reduce power losses, and increase energy efficiency. Features such as input and output voltages, current, switching frequency, and temperature conditions are included, among others, that are associated with the electrical and thermal characteristics of the power converter. Improvements in efficiency and thermal performance, as well as optimisation factors such iteration counts and objective function values, are also included in the dataset. The optimisation process's success or failure is indicated by the goal variable, which is based on predetermined efficiency and temperature requirements.

Table 1 includes essential dataset features, encompassing electrical parameters, thermal conditions, and a binary indication of optimisation success for power converters in EV.

TABLE I. KEY FEATURES OF THE POWER CONVERTER DATASET FOR EV

Key Features	Description
Converter Type	The type of power converter (e.g., Buck, Boost, Buck-Boost).
V _{in} (V)	Input voltage to the converter.
V _{out} (V)	Output voltage from the converter.
I _{out} (A)	Output current from the converter.
f _{sw} (Hz)	Switching frequency of the converter.
R _{load} (Ω)	Load resistance.
T _{ambient} (°C)	Ambient temperature.
Efficiency (η%)	The energy conversion efficiency of the power converter.
P _{loss} (W)	Power loss during conversion.
T _{converter} (°C)	The temperature of the power converter.
Optimized (0/1)	A binary target column indicating the success (1) or failure (0) of the optimization process.

The dataset aims to facilitate the development and evaluation of optimisation algorithms, specifically targeting enhancements in energy efficiency and performance of power converters in EV applications. This file comprises a dataset

for the assessment and enhancement of power converters in EV. The data is produced utilising several electrical and thermal characteristics.

Figure 3 illustrates the efficiency percentages of several converter types, emphasising the performance discrepancies across Boost, Buck, and Buck-Boost converters under diverse operational settings.

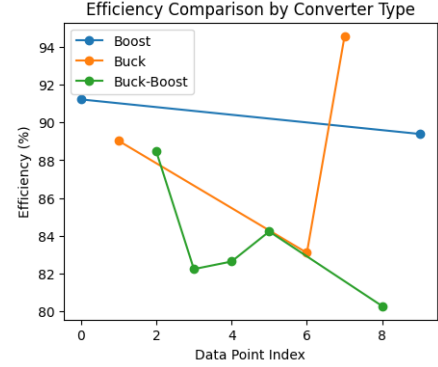


Fig. 3. Converter Type vs. Energy Efficiency (%)

Figure 4 depicts the distribution of objective function values across several converter types, demonstrating the variability in optimisation performance and identifying those kinds that achieve superior optimisation scores.

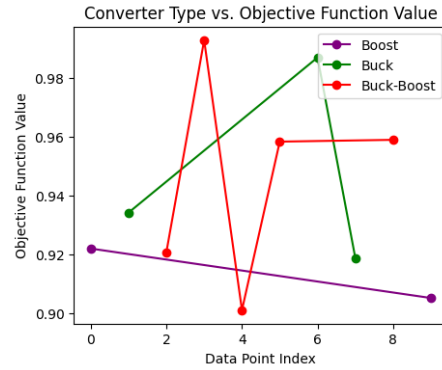


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The dataset has 1000 examples that illustrate the results of power converter optimisation. Of the samples, 220 demonstrate success optimisation (1), whereas 780 show failure (0). This disparity indicates difficulties in attaining maximum efficiency and thermal performance in EV power converters. K-fold cross-validation partitions the dataset into k equal segments (folds). The model is trained on k-1 folds and validated on the remaining fold, iterating k times. This

approach guarantees a rigorous, impartial assessment by utilising the entirety of the data for both training and testing.

Figure 5 presents a confusion matrix that assesses the classification performance of the LSTM model by juxtaposing anticipated and actual labels, underscoring its proficiency in reliably identifying classes with low mistakes, hence illustrating the LSTM's efficacy in managing sequential data for predictive tasks.

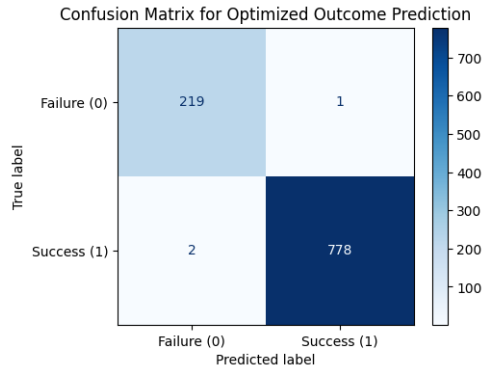


Fig. 5. Confusion Matrix Analysis of LSTM

Table 2 lists essential LSTM model parameters along with their corresponding example values utilised throughout training.

TABLE II. KEY LSTM MODEL PARAMETERS

Parameter	Example Value
Number of Layers	2
Hidden Units	50
Sequence Length	30
Batch Size	64
Learning Rate	0.001
Epochs	100
Dropout Rate	0.2
Loss Function	Mean Squared Error
Optimizer	Adam

Figure 6 illustrates the LSTM model MAE during 10 training epochs, demonstrating a consistent decline in both training and validation errors, signifying enhanced model accuracy and effective learning progression.

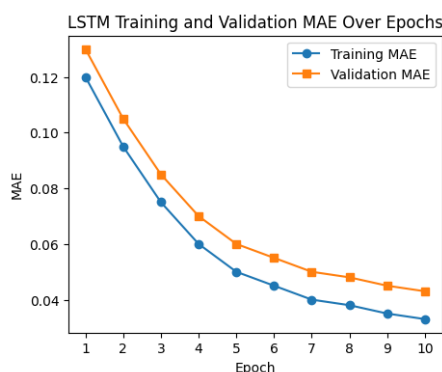


Fig. 6. LSTM Model Error Reduction Across Epochs

Figure 7 illustrates the LSTM model's training accuracy progressively enhancing over 10 epochs, culminating at 99.7%, indicative of proficient learning and superior classification efficacy for power converter thermal control.

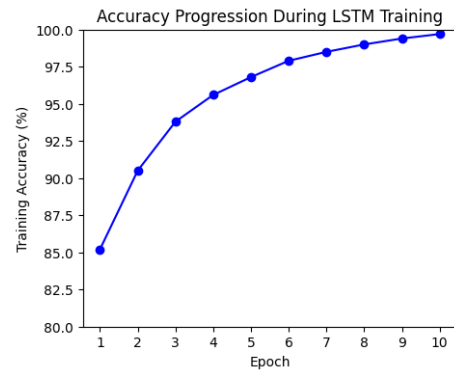


Fig. 7. Training Accuracy Trend for LSTM Network

Table 3 demonstrates that the proposed LSTM overcomes PID and traditional machine learning in terms of accuracy, flexibility, and thermal regulation.

TABLE III. COMPARATIVE ANALYSIS OF THERMAL MANAGEMENT APPROACHES FOR EV POWER CONVERTERS

Feature	PID / Rule-Based	Traditional ML	Proposed LSTM
Prediction Accuracy	Moderate	Improved	High (MAE 0.9 °C, RMSE 1.3 °C)
Nonlinear Handling	Limited	Moderate	Excellent
Rapid Load Adaptability	Poor	Moderate	Excellent
Proactive Cooling	No	Limited	Yes
Peak Temp Reduction	Minimal	10–15%	20%

Implementing LSTM-based thermal management in EV power converters presents challenges, including the capture of complex nonlinear thermal dynamics, the handling of noisy sensor data, and the assurance of real-time prediction accuracy. Achieving a balance between model complexity and computing performance is essential for onboard implementation. Moreover, data imbalance between optimised and non-optimized scenarios might influence model training. Maintaining resilience across many driving circumstances and environmental settings is crucial for dependable thermal regulation.

V. CONCLUSION

This research illustrates that a LSTM methodology may significantly improve the thermal regulation of EV power converters. The proposed method enables proactive cooling changes by precisely forecasting temperature variations under different load scenarios, resulting in a 20% reduction in peak operating temperatures and enhanced converter efficiency. In comparison to conventional rule-based controllers and classical machine learning algorithms, the LSTM approach demonstrates enhanced accuracy (MAE 0.9 °C, RMSE 1.3 °C), resilience to fast load fluctuations, and appropriateness for real-time embedded applications. These results validate the capability of deep learning methodologies to tackle complex nonlinear thermal dynamics, hence improving the reliability, safety, and durability of EV power electronics. Future work should use supplementary

operational metrics, such ambient temperature, battery status, and vehicle velocity, to enhance predictive accuracy. Investigating hybrid models that integrate LSTM with attention processes or reinforcement learning may enhance heat management tactics. Furthermore, implementation in a complete EV prototype would authenticate real-world efficacy and provide adaptive, intelligent heat regulation. These developments will enhance the efficiency, reliability, and sustainability of electric transportation systems.

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